



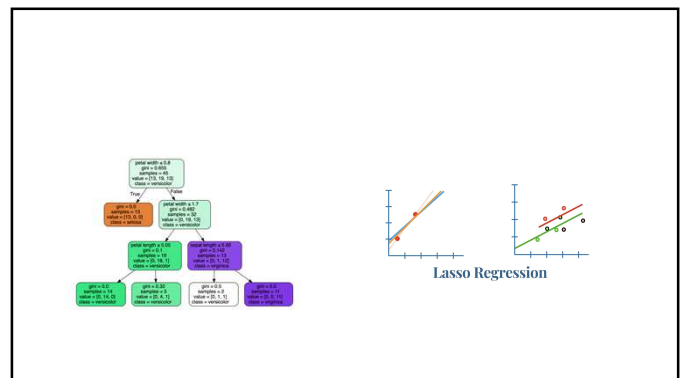
10.1. Terminologija i motivacija

- PRIMJERI:
 - Skaliranje i normalizacija
 - Popunjavanje vrijednosti
 - Odabir značajki
 - Kodiranje značajki
 - Konstrukcija značajke
 - Ekstrakcija značajki

10.2. Redukcija

- RAZLOZI REDUKCIJE:

- Redundancija
- Irelevantnost
- preopterećenosti



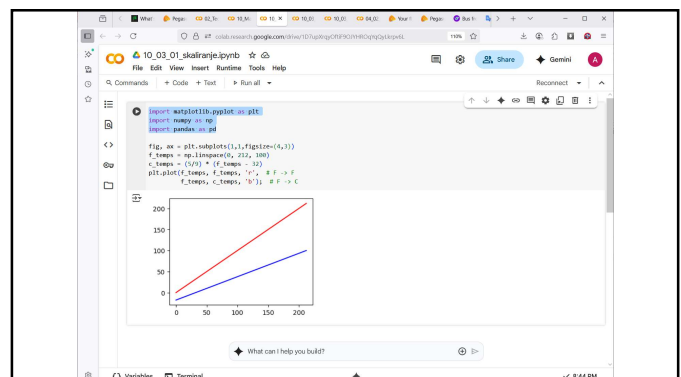
10.3. Skaliranje

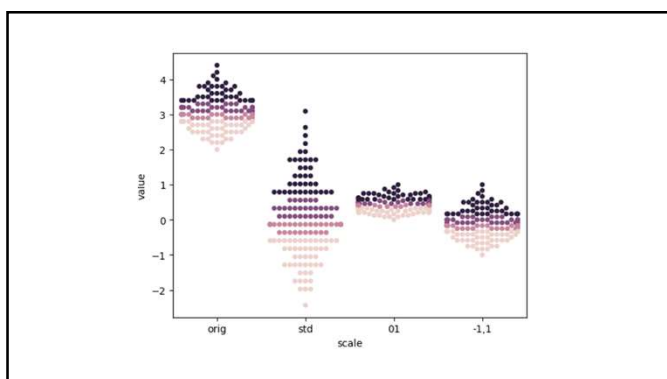
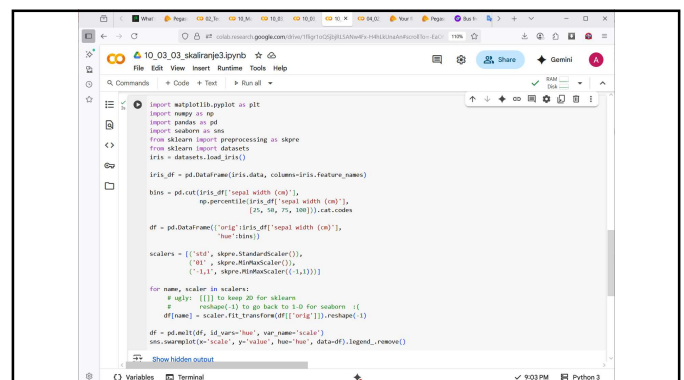
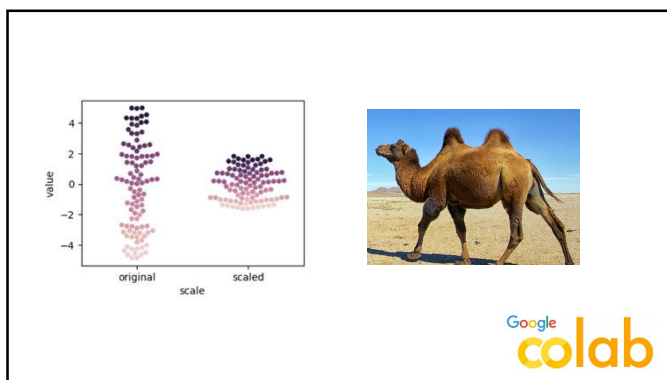
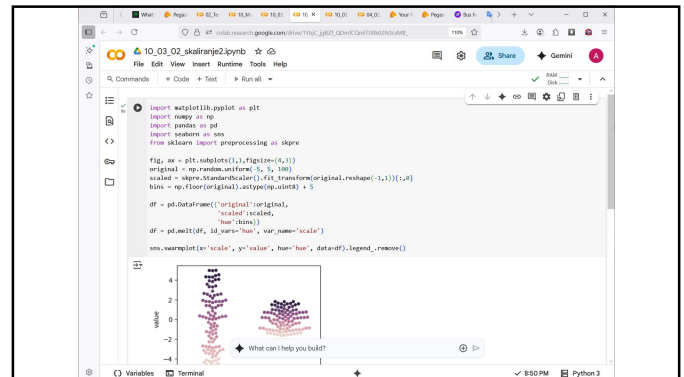
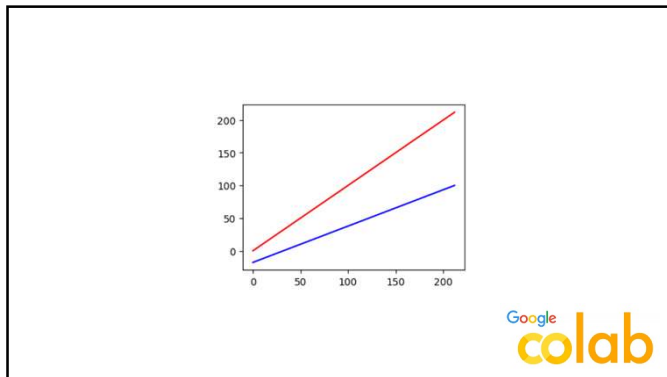
- skaliranje znači da prevodimo vrijednosti tako da su ekstremi različiti, a srednje vrijednosti pomaknute na neki dosljedan način

$$C = \frac{5(F-32)}{9} \quad C = \frac{5}{9}F - 17.7$$

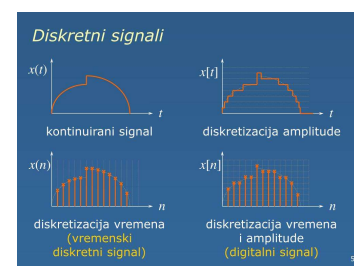
$$y = mx + b$$

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10.4 Diskretizacija



```

import pandas as pd
from sklearn import datasets
iris = datasets.load_iris()

iris_df = pd.DataFrame(iris.data, columns=iris.feature_names)
iris_df['species'] = iris.target_names[iris.target]
display(iris_df.iloc[[0,50,100]])

iris_df

```

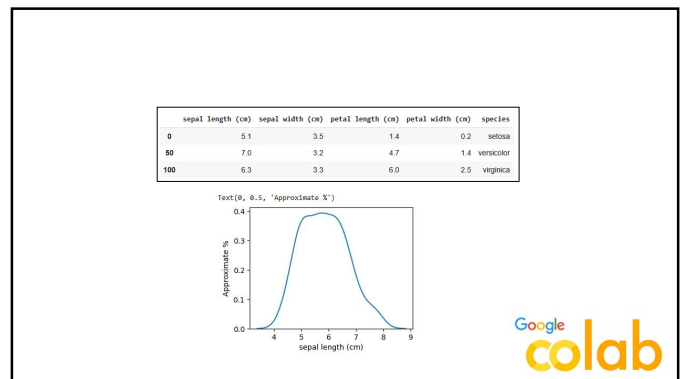
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
50	7.0	3.2	4.7	1.4	versicolor
100	6.3	3.3	6.0	2.5	virginica

```

import seaborn as sns
import matplotlib.pyplot as plt
plt.subplots(1,1,figsize=(4,4))
ax = sns.kdeplot(iris_df, x='sepal length (cm)')
ax.set_ylabel('Approximate %')

Text(0, 0.5, 'Approximate %')

```



```

import pandas as pd
from sklearn import datasets
from sklearn import preprocessing as skpre
import numpy as np

iris = datasets.load_iris()
iris_df = pd.DataFrame(iris.data, columns=iris.feature_names)

column = iris_df[['sepal length (cm)']]
col_mean = column.mean().values

both = column.copy()
both['> Mean'] = skpre.binarize(column, threshold=col_mean[0]).astype(np.bool_)

print('column Mean:', col_mean)
display(both.iloc[[0,50,100]])

Prikazi skrivene izlazne podatke

```

```

sep_len_series = iris_df['sepal length (cm)']
breaks = [sep_len_series.mean(),
           sep_len_series.max()]

```

Column Mean: [5.84333333]

sepal length (cm) > Mean		
0	5.1	False
50	7.0	True
100	6.3	True

10.5 Kategorijsko kodiranje



```

import pandas as pd
from sklearn import datasets

iris = datasets.load_iris()

iris_df = pd.DataFrame(iris.data, columns=iris.feature_names)
iris_df['species'] = iris.target_names[iris.target]

new_iris_df = pd.DataFrame(iris_df, columns=['petal length (cm)',
                                             'petal width (cm)',
                                             'species'])

new_iris_df.columns = pd.MultiIndex.from_tuples([('input ftr', 'target ftr'),
                                                  (new_iris_df.columns,
                                                   [1, 0, 0], [0, 1, 0], [0, 0, 1])])

new_iris_df.sort_index(axis='columns', inplace=True)
display(new_iris_df.iloc[[0,50,100]])

input ftr      target ftr
petal width (cm) species petal length (cm)

```

input ftrs		target ftr	
petal width (cm)	species	petal length (cm)	
0	0.2 setosa	1.4	
50	1.4 versicolor	4.7	
100	2.5 virginica	6.0	

- One-hot codiranje - Prevodi jedan stupac s više vrijednosti u više stupaca s jednom i samo jednom vrijednošću .



```
from sklearn import preprocessing as skpre
print("Numerical categories:")
Iris.target[[0, 50, 100]], sep='\n')

sparse = skpre.OneHotEncoder().fit_transform(Iris.target.reshape(-1,1))

print("One-hot coding:", sparse[[0,50,100]].todense(), sep='\n')
```

Numerical categories:
[0 1 2]
One-hot coding:
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]

Numerical categories:
[0 1 2]
One-hot coding:
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]



```
import pandas as pd
from sklearn import datasets
Iris = datasets.load_iris()

Iris_df = pd.DataFrame(Iris.data, columns=Iris.feature_names)
Iris_df['species'] = Iris.target_names[Iris.target]
encoded = pd.get_dummies(Iris_df, prefix='is')
encoded.iloc[[0,50,100]]
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	is_setosa	is_versicolor	is_virginica
0	5.1	3.5	1.4	0.2	True	False	False
50	7.0	3.2	4.7	1.4	False	True	False
100	6.3	3.3	6.0	2.5	False	False	True

```
# splicing dataframes together by merging
# recall 'Iris.target' is in terms of 0,1,2 not symbolic setosa, etc.
encoded_species = pd.get_dummies(Iris.target)
encoded_df = pd.merge(Iris_df, encoded_species,
                      right_index=True, left_index=True)
encoded_df.iloc[[0,50,100]]
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	0	1	2
0	5.1	3.5	1.4	0.2	setosa	True	False	False
50	7.0	3.2	4.7	1.4	versicolor	False	True	False
100	6.3	3.3	6.0	2.5	virginica	False	False	True

ohe = OneHotEncoder(drop='first',
sparse_output=False)

- Zašto izbacujemo jedan stupac kod OneHotEncoder ?
- MULTIKOLINEARNOST !

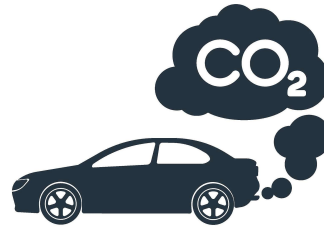


Primjer: DIJAMANTI



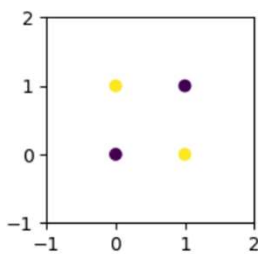
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Primjer: CO₂



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10.6 Odnosi i interakcije



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```
[1] In
import pandas as pd
var_data = [[0,0,0],
            [0,1,1],
            [1,0,1],
            [1,1,0]]
var_df = pd.DataFrame(var_data,
                      columns=['x1','x2','tgt'])
display(var_df)
```

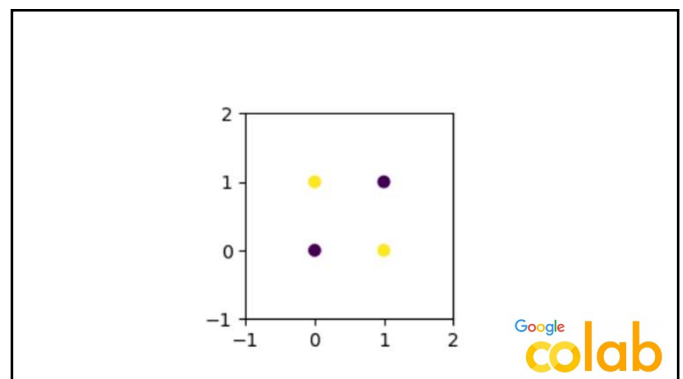
x1	x2	tgt
0	0	0
1	0	1
0	1	1
1	1	0

```
[2] In
from sklearn import linear_model
model = linear_model.LogisticRegression().fit(var_df[['x1', 'x2']],
                                              var_df['tgt'])
model.predict(var_df[['x1', 'x2']])
```

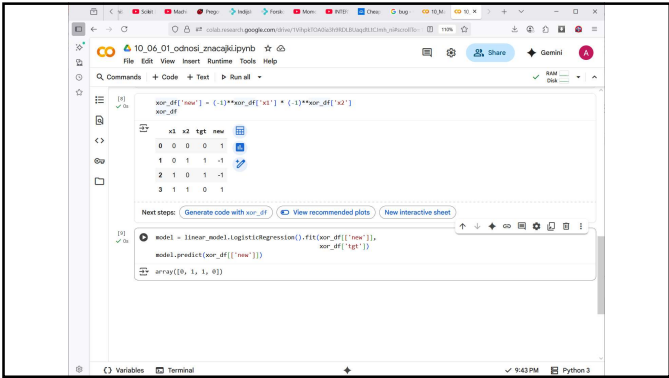
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```
[1] In
import matplotlib.pyplot as plt

fig, ax = plt.subplots(1,1,figsize=(2,2))
ax.scatter('x1', 'x2', data=var_df, c='tgt')
ax.set_xlim(-1, 2)
ax.set_ylim(-1, 2)
```



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colab



x1	x2	tgt	new
0	0	0	1
1	0	1	-1
2	1	0	-1
3	1	1	1